

CMSC 478: Assignment 1

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Part B: Explain concepts (simple + aligned with what you just did)

Write 2–4 sentences each, in your own words, and tie each explanation to the code you ran:

1. Why assert versions? What can break if library versions change?

When libraries are updated, some functionality might behave differently or been removed entirely. The reason why to assert versions is to make sure that the program is running stable versions of the libraries.

2. `pivot()`: What does pivoting do to the OECD table, and why is it useful here?

`pivot()` made it possible to organize the table by making each country as the index(rows) and their indicators as columns. It is useful here because it helps grab necessary columns such as the 'Life Satisfaction' column.

3. `merge()` on index: What does joining on country index accomplish?

There are two tables which are `oecd_bli` and `gdp_per_capita` and they share a common index which are the names of different countries. `merge()` made it possible to combine each country's indicator and GDP per capita.

4. Why `np.c_[ ]`? Explain how it converts a Pandas column into a 2D feature matrix.

`scikit-learn`'s `fit()` method needs a 2D array format. The `np.c_[ ]` converts a 1D array into 2D array by arranging the values in a single column 2D array.

5. Scatter plot: What is each dot? What pattern suggests a linear model might work?

Each dot represents each country where the x-axis is the 'GDP per capita' and the y-axis is the 'Life Satisfaction.' The graph goes upward from left to right which suggests a linear model might work.

6. Linear Regression (`fit` / `predict`): In plain language, what does `fit()` learn, and what does `predict()` do?

The `fit()` method finds the slope and intercept of the line that minimizes the error. The `predict()` method outputs an estimated life satisfaction value and plugging it into the line equation.

7. Data cleaning via indices: Why might the notebook remove a few countries (outliers/missingness)? What is the risk of manual removal?

Some countries may not have accurate or missing data which could affect the result of the prediction. The risk of manual removal could create a controlled outcome or the possibility of inaccurate results.

Part C: Interpret results (what the model is telling you)

Answer the following, referencing your plot and output:

1. Interpret the trend: Describe the relationship between GDP per capita and life satisfaction in this sample.

The general trend is that as GDP per capita increases the life satisfaction increases as well.

2. What does the Cyprus prediction mean? Explain what 5.96 represents (units/scale) and what assumptions are embedded.

The Cyprus prediction means that the life satisfaction is 5.96 that is with the scale 0-10 where 0 is the worst and 10 being the best. Based on the graph, the predicted result for Cyprus falls a little below the middle. The assumptions embedded are no other factors were considered such as cultural and political factors.

3. Limitations: Give three reasons this model could be misleading (e.g., correlation vs causation, omitted variables, sample bias, nonlinearity/outliers). This model could be misleading because it could correlate that wealth causes happiness. Additionally, there are also other factors that drive happiness other than GDP that the model did not take into consideration. Also, the dataset only included OECD countries so that could possibly result with a biased data.

4. Generalization: Would you trust this model to predict life satisfaction for a country with GDP per capita far outside the observed range? Why?

No, I would not trust this model to predict life satisfaction for a country with GDP per capita that is far outside the observed range. Since it is a linear model, it assumes that when GDP increases so does life satisfaction. Any extremes far outside the observed range could potentially create unreliable predictions because there may be diminishing returns once a certain GDP is reached.

Part D: Gemini question: Would Google Gemini generate a similar result?

Answer with a short, precise argument:

Gemini would not generate result because it could preprocess the the data differently, use a different model other than linear regression, and different versions of the libraries. Gemini could organize the data differently and remove more or less outliers. Also, Gemini might decide to use another model that it

deemed better because it produced a slightly different graph after preprocessing the data . Finally, Gemini could use a recent updated versions of the libraries which could produce a different code.